Functional Analysis — Homework 6

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Problem 1. Let $f : \mathbb{C} \to \mathbb{C}$ be an analytic function, and let

$$f(x, y) = u(x, y) + iv(x, y)$$

be its decomposition into real and imaginary parts. Consider the limit definition of the complex derivative at a point $z = x + iy$:

$$f'(z) = \lim_{w \to 0} \frac{f(z + w) - f(z)}{w}.$$

(a) Evaluate this limit by letting $w \to 0$ strictly along the real axis. Express the result in terms of the partial derivatives of u and v.

(b) Now evaluate the same limit by letting $w \to 0$ strictly along the imaginary axis.

Since f is analytic, the limit must be independent of the path. Equate your results from (a) and (b) to prove the Cauchy-Riemann equations:

$$u_x = v_y, \quad u_y = -v_x.$$

Proof.

(a) Let $z = (x, y)$ and $w = (w_x, w_y)$, where in this case $w_y = 0$ because we are taking the limit along the real line. Then

$$\begin{aligned}
f'(z) &= \lim_{w \to 0} \frac{f((x, y) + (w_x, w_y)) - f(x, y)}{w_x + iw_y} \\
&= \lim_{w_x \to 0} \frac{f(x + w_x, y) - f(x, y)}{w_x} \\
&= \lim_{w_x \to 0} \frac{u(x + w_x, y) + iv(x + w_x, y) - u(x, y) - iv(x, y)}{w_x} \\
&= \lim_{w_x \to 0} \frac{u(x + w_x, y) - u(x, y)}{w_x} + i \lim_{w_x \to 0} \frac{v(x + w_x, y) - v(x, y)}{w_x} \\
&= u_x + iv_x.
\end{aligned}$$

(b) Now if $w_x = 0 \neq w_y$, then

$$\begin{aligned}
f'(z) &= \lim_{w \to 0} \frac{f((x,y)+(w_x,w_y)) - f(x,y)}{iw_y} \\
&= \lim_{w_y \to 0} \frac{f(x, y+w_y) - f(x,y)}{iw_y} \\
&= \lim_{w_y \to 0} \frac{u(x,y+w_y) + iv(x,y+w_y) - u(x,y) - iv(x,y)}{iw_y} \\
&= \lim_{w_y \to 0} \frac{u(x,y+w_y) - u(x,y)}{iw_y} + i \lim_{w_y \to 0} \frac{v(x,y+w_y) - v(x,y)}{iw_y} \\
&= -i \lim_{w_y \to 0} \frac{u(x,y+w_y) - u(x,y)}{w_y} + \lim_{w_y \to 0} \frac{v(x,y+w_y) - v(x,y)}{w_y} \\
&= -iu_y + v_y.
\end{aligned}$$

Setting the two real and imaginary parts of the equations equal (respectively), we get $u_x = v_y$ and $v_x = -u_y$ as desired. □

Problem 2. Use the limit definition to show that $f(z) = \overline{z}$ is not complex differentiable anywhere, and $g(z) = |z|^2$ is not complex differentiable at any $z \neq 0$, but is complex differentiable at $z = 0$.

Proof. Taking the derivative along the real axis, we get

$$\begin{aligned}
f'(z) &= \lim_{w_x \to 0} \frac{f(x+w_x, y) - f(x,y)}{w_x} \\
&= \lim_{w_x \to 0} \frac{x + w_x - yi - (x - yi)}{w_x} \\
&= \lim_{w_x \to 0} \frac{x + w_x - yi - x + yi}{w_x} \\
&= \lim_{w_x \to 0} \frac{w_x}{w_x} \\
&= \lim_{w_x \to 0} 1 \\
&= 1,
\end{aligned}$$

and taking the derivative along the imaginary axis, we get

$$\begin{aligned} f'(z) &= \lim_{w_y \to 0} \frac{f(x, y+w_y) - f(x,y)}{w_y} \\ &= \lim_{w_y \to 0} \frac{x - i(y+w_y) - (x - iy)}{w_y} \\ &= \lim_{w_y \to 0} \frac{x - iy - iw_y - x + iy}{w_y} \\ &= \lim_{w_y \to 0} \frac{-iw_y}{w_y} \\ &= \lim_{w_y \to 0} -i \\ &= -i. \end{aligned}$$

Since $1 \neq -i$, f is not differentiable. □

Problem 3. Fix a small, positive radius $r < 1$ and let γ_1 and γ_2 be the positively oriented circles of radius r centered at $z = 1$ and $z = 2$, respectively:

$$\gamma_1 = 1 + re^{it}, \qquad \gamma_2(t) = 2 + re^{it}, \qquad t \in [0, 2\pi].$$

Compute the following line integrals for $j = 1, 2$:

$$(a) \int_{\gamma_j} \frac{dz}{1-z} \qquad (b) \int_{\gamma_j} \frac{dz}{2-z} \qquad (c) \int_{\gamma_j} \frac{dz}{(1-z)^2} \qquad (d) \int_{\gamma_j} \frac{dz}{(2-z)(1-z)}.$$

Hint: for item (d), partial fractions can immediately reduce it to (a) and (b).

Proof. (a) Note that $\int_{\gamma_j} \frac{dz}{1-z} = -\int_{\gamma_j} \frac{dz}{z-1}$. Since 1 is in the disk enclosed by γ_1 but not γ_2, we have

$$\begin{aligned} \int_{\gamma_1} \frac{dz}{1-z} &= -2\pi i \\ \int_{\gamma_2} \frac{dz}{1-z} &= 0 \end{aligned}$$

(b) By similar logic, we get

$$\begin{aligned} \int_{\gamma_1} \frac{dz}{2-z} &= 0 \\ \int_{\gamma_2} \frac{dz}{2-z} &= -2\pi i \end{aligned}$$

(c) Hmm not sure about this one. I feel like it should be this, but really just by geometric vibes, comparing it to (a):

$$\begin{aligned} \int_{\gamma_1} \frac{dz}{(1-z)^2} &= -2\pi i \\ \int_{\gamma_2} \frac{dz}{(1-z)^2} &= 0. \end{aligned}$$

The only other idea I have is that the first one should be positive because all the values are squared, but these are complex values we're talking about, so it could still be negative. NOTE: I now know that it should be zero for both γ_1 and γ_2, which I looked up so I could use it in the next problem.

(d) By partial fraction decomposition, we have

$$\frac{-1}{2-z} + \frac{1}{1-z} = \frac{1}{(2-z)(1-z)},$$

so

$$\int_{\gamma_j} \frac{dz}{(2-z)(1-z)} = \int_{\gamma_j} \frac{1}{1-z} - \int_{\gamma_j} \frac{1}{2-z}.$$

. Using (a) and (b) then, we get

$$\int_{\gamma_1} \frac{dz}{(2-z)(1-z)} = -2\pi i$$
$$\int_{\gamma_2} \frac{dz}{(2-z)(1-z)} = 2\pi i.$$

□

Problem 4. Consider the linear operator $A : \mathbb{C}^3 \to \mathbb{C}^3$ represented in the standard basis by the matrix

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Note that the spectrum of A is $\sigma(A) = \{1, 2\}$.

(a) Find the kernel of $(A-1)^2$ and the kernel of $A-2$. These are the generalized eigenspaces of A for the eigenvalues 1 and 2.

(b) Compute the resolvent matrix $R(z) = (A - z)^{-1}$ for $z \notin \sigma(A)$. Simplify the rational functions in each entry completely.

(c) For each eigenvalue $\lambda_j = j, j = 1, 2$ compute the operator-valued integral

$$P_j = -\frac{1}{2\pi i} \int_{\gamma_j} R(z) dz,$$

where γ_j is a small positively oriented circle enclosing only λ_j.

(d) Show that P_1 and P_2 are projections, that their images are the generalized eigenspaces of A,

$$\mathrm{Im}(P_1) = \ker(A-1)^2, \qquad \mathrm{Im}(P_2) = \ker(A-2),$$

and that they complement each other, meaning:

$$P_1 + P_2 = 1, \qquad P_1 P_2 = 0.$$

P_1 and P_2 are called *Riesz projections* associated with A.

Proof.

(a) Note that $(A-1)^2 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and $A - 2 = \begin{pmatrix} 0 & 1 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{pmatrix}$. Multiplying by generic elements, we get

$$\begin{aligned} (A-1)^2 \begin{pmatrix} a \\ b \\ c \end{pmatrix} &= \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} \\ &= \begin{pmatrix} a+b \\ 0 \\ 0 \end{pmatrix} \\ (A-2) \begin{pmatrix} a \\ b \\ c \end{pmatrix} &= \begin{pmatrix} 0 & 1 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} \\ &= \begin{pmatrix} b-c \\ c-b \\ -c \end{pmatrix}, \end{aligned}$$

so if we set each one equal to 0, we get in the first case $\{(a, -a, c)\}$, and in the second case $\{(a, 0, 0)\}$ for the kernels.

(b) After lots of algebra, we get this:

$$R(z) = \begin{pmatrix} \frac{1}{2-z} & -\frac{1}{(2-z)(1-z)} & \frac{1}{(1-z)^2} \\ 0 & \frac{1}{1-z} & -\frac{1}{(1-z)^2} \\ 0 & 0 & \frac{1}{1-z} \end{pmatrix}$$

(c) Using Problem 3, we get

$$
\begin{aligned}
P_1 &= -\frac{1}{2\pi i}\int_{\gamma_1} R(z)dz \\
&= -\frac{1}{2\pi i}\int_{\gamma_1} \begin{pmatrix} \frac{1}{2-z} & -\frac{1}{(2-z)(1-z)} & \frac{1}{(1-z)^2} \\ 0 & \frac{1}{1-z} & -\frac{1}{(1-z)^2} \\ 0 & 0 & \frac{1}{1-z} \end{pmatrix} dz \\
&= -\frac{1}{2\pi i}\begin{pmatrix} \int_{\gamma_1}\frac{1}{2-z} & -\int_{\gamma_1}\frac{1}{(2-z)(1-z)} & \int_{\gamma_1}\frac{1}{(1-z)^2} \\ 0 & \int_{\gamma_1}\frac{1}{1-z} & -\int_{\gamma_1}\frac{1}{(1-z)^2} \\ 0 & 0 & \int_{\gamma_1}\frac{1}{1-z} \end{pmatrix} \\
&= -\frac{1}{2\pi i}\begin{pmatrix} 0 & 2\pi i & 0 \\ 0 & -2\pi i & 0 \\ 0 & 0 & -2\pi i \end{pmatrix} \\
&= \begin{pmatrix} 0 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
P_2 &= -\frac{1}{2\pi i}\int_{\gamma_2} R(z)dz \\
&= -\frac{1}{2\pi i}\int_{\gamma_2} \begin{pmatrix} \frac{1}{2-z} & -\frac{1}{(2-z)(1-z)} & \frac{1}{(1-z)^2} \\ 0 & \frac{1}{1-z} & -\frac{1}{(1-z)^2} \\ 0 & 0 & \frac{1}{1-z} \end{pmatrix} dz \\
&= -\frac{1}{2\pi i}\begin{pmatrix} \int_{\gamma_2}\frac{1}{2-z} & -\int_{\gamma_2}\frac{1}{(2-z)(1-z)} & \int_{\gamma_2}\frac{1}{(1-z)^2} \\ 0 & \int_{\gamma_2}\frac{1}{1-z} & -\int_{\gamma_2}\frac{1}{(1-z)^2} \\ 0 & 0 & \int_{\gamma_2}\frac{1}{1-z} \end{pmatrix} \\
&= -\frac{1}{2\pi i}\begin{pmatrix} -2\pi i & -2\pi i & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}
\end{aligned}
$$

(d) We have

$$
\begin{aligned}
P_1^2 &= \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}^2 = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \\
P_2^2 &= \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}
\end{aligned}
$$

and

$$P_1 \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} a-b \\ 0 \\ 0 \end{pmatrix}$$
$$P_2 \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} b \\ b \\ c \end{pmatrix},$$

which when compared to the calculations from earlier doesn't seem to quite match up. That they complement each other is apparent from the matrix representations, and

$$\begin{aligned} P_1 P_2 &= \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

as desired.

□